Zakaria Abouhammadi | Machine Learning Engineer

Computer Science Student / Applied AI

– Valencia, Spain

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Professional Profile

Machine Learning Engineer intern at **Solver Intelligent Analytics**, currently designing and building the document generation subsystem for the Comú d'Encamp (Andorra) project, using a **LangGraph** pipeline and **python-docx** on top of a self-hosted vLLM instance.

Final-year Computer Science student specialized in **AI systems**, **advanced RAG** and **backend development**. Hands-on experience building **RAG pipelines** (HyDE, MultiQuery, rerankers, Chroma Cloud), **agents** with LangChain/LangGraph/CrewAI/AutoGen and **Transformer** and **CNN** models from scratch. Designed multi-agent debate systems, implemented a *Vision Transformer*, and developed scalable web applications. Advocate of **Scrum**, with experience as developer and Scrum Master, delivering robust and well-documented solutions.

Experience

Solver Intelligent Analytics — **Valencia, Spain**

Machine Learning Engineer, Internship contract · Hybrid · 4 months — *Jan 2026 – Present*

Built an AI assistant that automatically generates public procurement documents for the municipality of Encamp (Andorra). Through a guided conversation with the user, the system collects the required information and produces the final Word document, covering twelve different variants depending on the type of administrative procedure. Responsible for the design and implementation of the full generation module: a data collection and validation pipeline with **LangGraph**, a rendering engine based on Word templates using **python-docx**, and integration with the client's main multilingual chatbot. Additional work on prompt engineering to adapt the LLM to administrative Catalan.

Stack: Python, LangGraph, python-docx, FastAPI, Docker, vLLM (Qwen3-30B), RAG, prompt engineering.

Education

Universitat de València — **Valencia, Spain**

Bachelor's Degree in Computer Science Engineering — *Sep 2022 – Sep 2026 (expected)*

Key subjects: Data structures and algorithms, Web application development, Intelligent systems, Software engineering, Databases.

Technical Skills

Languages: Python, C/C++, Java, JavaScript, HTML5, CSS3, SQL

AI / ML: PyTorch, TensorFlow, Keras, scikit-learn, Transformers, BERT/ViT, LangChain, LangGraph, CrewAI, AutoGen, BeeAI, Ollama, vLLM

RAG / NLP: ChromaDB (Cloud), FAISS, BM25, HyDE, MultiQuery, Reranking, Retrieval Evaluation, Prompt Engineering, Knowledge Graphs

Backend: FastAPI, Flask, Java EE (JSP/Servlets), REST APIs, python-docx

Data: PostgreSQL, MySQL, SQLite, MongoDB, pandas, NumPy

DevOps: Docker, Docker Compose, Git & GitHub Actions, Google Cloud, Linux

Methodologies: Scrum *(developer and Scrum Master)*

Highlighted Skills

Advanced RAG (HyDE, MultiQuery, rerankers) · Vector databases (Chroma, FAISS) · Agents (LangGraph, CrewAI, AutoGen, BeeAI) · Transformers (BERT, ViT) · Convolutional networks · Risk modeling and forecasting · Decision trees · Image analysis · EDA & Feature engineering · Optimization & hyperparameters · PyTorch fundamentals · Document generation with templates

Soft Skills

Multicultural teamwork · Trilingual communication (ES/EN/FR) · Analytical thinking · Time management · Continuous learning · Academic leadership

Certifications & Courses

Specializations & Programs

Nov 2025: **PyTorch for Deep Learning (Professional Certificate) — DeepLearning.AI**
ID: ZYEUD91L9FYS · Deep learning with PyTorch · Transfer learning for CV & NLP · Model optimization and deployment

Oct 2025: **Building AI Agents and Agentic Workflows (Specialization) — IBM**
ID: UROX1SUIDGWA · Design of agentic workflows, tool use, orchestration and evaluation

Oct 2025: **AI for Medicine Specialization — DeepLearning.AI**
ID: 6MOISRA2N5WK · Precision medicine, image analysis, risk modeling, decision trees, forecasting

Sep 2025: **RAG for Generative AI Applications (Specialization) — IBM**
ID: W2UQZCZMJHKV · End-to-end RAG system design and deployment

Feb 2025: **Deep Learning Specialization — DeepLearning.AI**
ID: EOAJ4XETDWMR · Neural networks, CNN, optimization, regularization, sequence models

Nov 2024: **Machine Learning — DeepLearning.AI (Andrew Ng)**
ID: 9PL5C8QDGYQ6 · Supervised, unsupervised, regularization, evaluation

Nov 2024: **IBM Introduction to Machine Learning — IBM**
ID: 59X2YJWXX9SS · Foundations of ML, supervised and unsupervised learning, model evaluation

RAG & Agents

Sep 2025: **Advanced RAG with Vector Databases and Retrievers — IBM**
ID: TSXE4ZG7MDFQ

Sep 2025: **Agentic AI with LangGraph, CrewAI, AutoGen and BeeAI — IBM**
ID: AWXAVIH36K1F

Sep 2025: **IBM RAG and Agentic AI — IBM**
ID: X3774PPHATF6

Jul 2025: **Build RAG Applications: Get Started — IBM**
ID: 596YXNIVML4A

Jul 2025: **Fundamentals of Building AI Agents — IBM**
ID: ZGQWDFK0VO5H

Jul 2025: **Vector Databases for RAG: An Introduction — IBM**
ID: K46CQ74O5CKA

Jul 2025: **AI Enhancement with Knowledge Graphs — Mastering RAG Systems — Packt**
ID: 28DS7P6LJIE8

Jul 2025: **Agentic AI with LangChain and LangGraph — IBM**
ID: V82LDS06UQN4

Jul 2025: **Develop Generative AI Applications: Get Started — IBM**
ID: 2EPBGK0YLTIP

Jul 2025: **Build Multimodal Generative AI Applications — IBM**
ID: IW6XFBO3EGSN

PyTorch & Deep Learning

Nov 2025: **PyTorch: Fundamentals — DeepLearning.AI**
ID: 6PJ2XY48LZZO · PyTorch basics, Tensors and Autograd

Nov 2025: **PyTorch: Techniques and Ecosystem Tools — DeepLearning.AI**
ID: CR0RUHDBQ5CS · Torch.Optim, PyTorch Lightning, ecosystem tools

Nov 2025: **PyTorch: Advanced Architectures and Deployment — DeepLearning.AI**
ID: CK4L1IG1BJ5L · Encoder–Decoder, ResNet, DenseNet, deployment

Sep 2025: **Building and Training Neural Networks with PyTorch — Packt**
ID: N4BCRQSYJ7D2 · CNNs

Sep 2025: **Foundations and Core Concepts of PyTorch — Packt**
ID: OJG0HNZ0ZQC9 · ML fundamentals

NLP & Transformers

Mar 2025: **Transformer Models and BERT Model — Google Cloud**
ID: K2LDLD7XJIM6

Feb 2025: **Attention Mechanism — Google Cloud**
ID: LJPL5IN4LHON

ML/DL Fundamentals

2024–2025: **Convolutional Neural Networks**
ID: Y4QTHRNL3HE7 · TensorFlow, CNN, facial recognition, object detection & segmentation

2024–2025: **Exploratory Data Analysis for Machine Learning**
ID: 8PVON5MQJ62R · EDA, feature engineering, statistical tests

2024–2025: **Improving Deep Neural Networks**
ID: CNCPQJS4KTFE · Regularization, optimization, hyperparameters

2024–2025: **Neural Networks and Deep Learning**
ID: KNJNB8WQWAJP

2024–2025: **Structuring Machine Learning Projects**
ID: Z3KQ6DQXD2GM

2024–2025: **Supervised Machine Learning: Classification**
ID: HYIO24EKA6SX

2024–2025: **Supervised Machine Learning: Regression**
ID: LDQ28KIIHHG7

2024–2025: **Unsupervised Machine Learning**
ID: 52M2R676600S

Nov 2024: **Supervised Machine Learning: Classification (Badge)**
Digital certificate

Web, Cloud & Networks

Feb 2025: **Google AI Essentials — Google**
ID: X51IOTBS2BSE

Feb 2025: **Advanced Styling with Responsive Design — Univ. Michigan**
ID: VHSLGPRZGP26

Feb 2025: **Interactivity with JavaScript — Univ. Michigan**
ID: JODZD9YLKDBT

Feb 2025: **Introduction to CSS3 — Univ. Michigan**
ID: FDU5K7VCNMBP

Feb 2025: **Introduction to HTML5 — Univ. Michigan**
ID: B9898ISXJUTR

Feb 2025: **Introduction to Networking — NVIDIA**
ID: AUQMCFVGVRP0

Feb 2025: **AI Infrastructure and Operations Fundamentals — NVIDIA**
ID: NFNQ1HP3ODGH

Projects

: **asistente-uv** (Bachelor's Thesis)

LangGraph, Ollama, ChromaDB, BM25, Playwright, FastAPI, React, Docker

RAG assistant that answers questions about the University of Valencia using official information extracted from *uv.es*. Combines a local document store with real-time web search when information is missing, and grows automatically with each new query. ReAct-style agent on LangGraph with hybrid retrieval (BM25 + dense + RRF) and Playwright-based scraping.

: **CodeScribe**

Python, LangGraph, Ollama, Click, Rich, Pydantic, uv

CLI tool that scans an entire project and adds documentation and comments to every source file using a local LLM via Ollama. Deterministic LangGraph pipeline: scan → classify → chunk → LLM → post-process → write back. Supports dry-run with diff, backups, interactive menu and automatic README generation.

: **chatbot — LangGraph RAG Assistant**

LangGraph, BM25, Dense Embeddings, RRF, SQLite, Selenium

Conversational QA system with hybrid retrieval fused via Reciprocal Rank Fusion, strict chunk-level citations, adaptive Selenium-based web enrichment and short/long-term memory backed by SQLite. Orchestrated in a single LangGraph instance with clearly contracted nodes. End-to-end multilingual.

: **llama_4_scratch**

PyTorch, NumPy, Jupyter

From-scratch implementation and step-by-step explanation of the key components of a LLaMA-style **decoder Transformer**: tokenizer, scaled dot-product attention and Multi-Head Latent Attention. Every operation commented with numerical examples and visualizations.

: **multiquery-hyde-rag-chroma-cloud**

Python, RAG, LangChain, Chroma Cloud

Advanced RAG pipeline with **HyDE** and **MultiQuery** to expand coverage, indexing in **Chroma Cloud**, and reranking to improve response accuracy; includes retrieval evaluation and reproducible documentation.

: **advanced-hyde-rag**

Python, RAG, LangChain

RAG implementation with HyDE and synthetic query generation, chunking and embeddings handling, comparison of retrieval strategies and latency analysis.

: **ranked-rag-pipeline**

Python, Cross-Encoder, Sentence Transformers

Retrieval chain with **rerankers** and filters, coverage/precision metrics and utilities for source traceability.

: **bm25-dense-retrieval**

BM25, Embeddings, RRF

Practical study of hybrid retrieval combining BM25 (lexical) with dense embeddings (semantic) and Reciprocal Rank Fusion. Performance comparison across query types.

: **data-chat**

LangChain, LLMs, SQL

Exploration of **text-to-SQL** and data chat patterns with agents: translate natural language questions into structured queries, execute them and return summarized answers.

: **MultiagentDebate**

Python, LangChain/LangGraph

Multi-agent debate system (scientific, economic, historical, psychological, refutation) with supervisors and automatic evaluation; orchestration with LangGraph and evidence logging.

: **VisionTransformer**

PyTorch, Computer Vision

Educational implementation of a Vision Transformer (ViT): patch embedding, multi-head attention, MLP head and training on vision datasets, with comparative analysis against CNN architectures.

: PoetryTransformerPredict
PyTorch, NLP
Bigram Transformer built from scratch (~9.5k embedding, 4 attention blocks) to generate Spanish poetry; training and sampling example.

: GiveMeSomeCredit — Default Risk
PyTorch, scikit-learn, pandas, MLP, AMP
Clean, reproducible baseline to predict serious delinquency within two years with PyTorch. Data audit, median imputation, quantile clipping, log1p transforms, RobustScaler, MLP with BatchNorm and GELU, mixed-precision training and early stopping on ROC-AUC.

: Recommender Systems
Surprise, NumPy, Collaborative Filtering
Collaborative filtering and content-based models, metrics (RMSE, MAP@K) and cross-validation; explanatory notebooks.

: Brain Tumor Classification
TensorFlow, VGG19
Brain tumor vs. non-tumor classification through fine-tuning of VGG19 and data augmentation; validation set evaluation.

: Amazigh Alphabets Recognition
TensorFlow, Keras
CNN network for tifinagh handwritten character recognition with $> 95\%$ accuracy; preprocessing and training pipeline.

: Handwritten Alphabets & Digit Recognition
TensorFlow, Keras, MNIST
CNN models for handwritten Latin letters and digits (MNIST), comparison with SVM/MLP and error analysis.

: Housing Price Prediction
Python, scikit-learn
Linear regression and Random Forest for housing price estimation; RMSE and R^2 comparison.

: TiendaWeb (Klawz)
Java/JSP, MySQL
Complete e-commerce with catalog, cart and payment gateway; MVC architecture, sessions and basic security.

: PhongIllumination
C, OpenGL, OpenMP, MPI
Implementation of the Phong illumination model in sequential, shared-memory and distributed parallel versions; speedup measurement.

: minilang
Python, PLY
Educational compiler for MiniLang with lexical, syntactic and semantic analysis, AST generation and test execution.

: ISII_AdministrarHospitales
Java, Full-stack
Hospital management system: medical records, inventory and access control; authentication and roles.

Languages

Berber: Native
Arabic: Native
Spanish: B2–C1 — *Full professional proficiency*
English: B2–C1 — *Full professional proficiency*
French: B2 — *Professional working proficiency*

Interests

Generative AI · Machine Learning · NLP · RAG · Databases · Compilers · Cloud Computing · Cybersecurity · Robotics · IoT · Video game development · Open Source · Optimization · Computer Vision · Drone photography · Hiking · Virtual/augmented reality